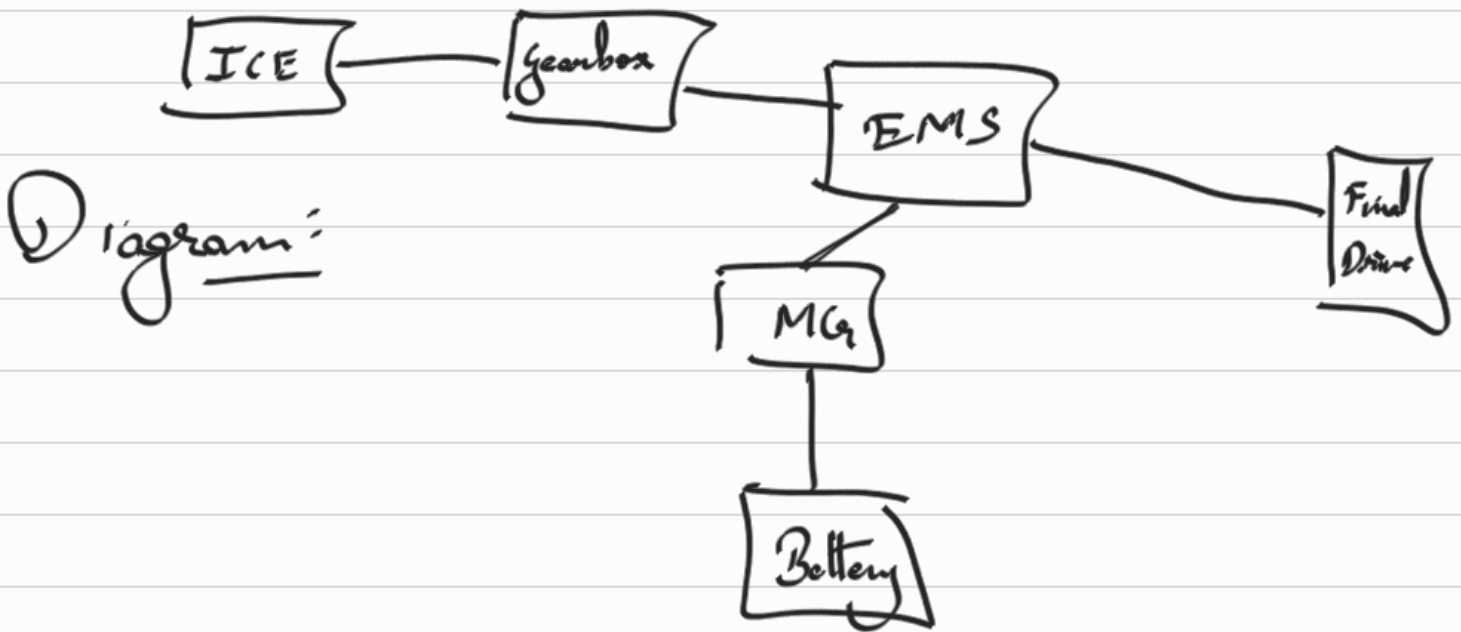


PHEV, Power Split Method.

Proof - of - Concept.

Efficient $\rightarrow$ Less emission (Esp. Urban Areas).

Cruising - ICE Efficiency $\uparrow$



Flow of Energy: Plug-in Power-Split PHEV

1. Stop & go - ($< 20\%$ SOC) EV only mode.

Heat & Hotter + Emission

High (Urban)
High Population Density

+

Consumers
Saving fuel Ec. Petrol

Gov.

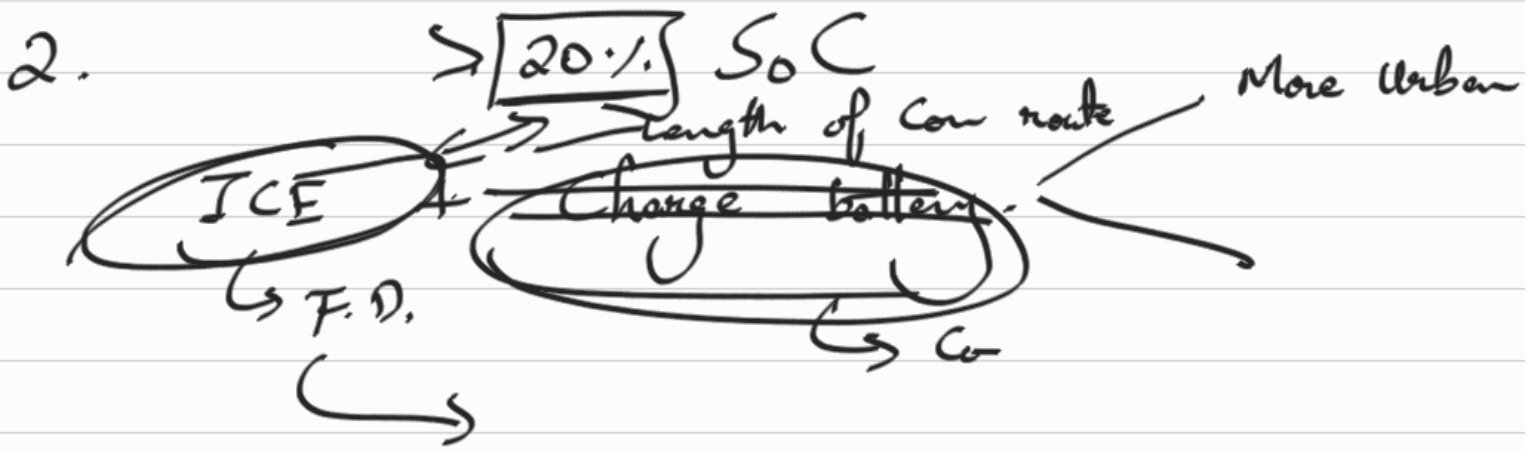
Try to have Max SOC
during Urban transit.

Traffic
Speed lim.
Avg.

GPS, Self-Drive.

Model AD, (Local Emission) $>$

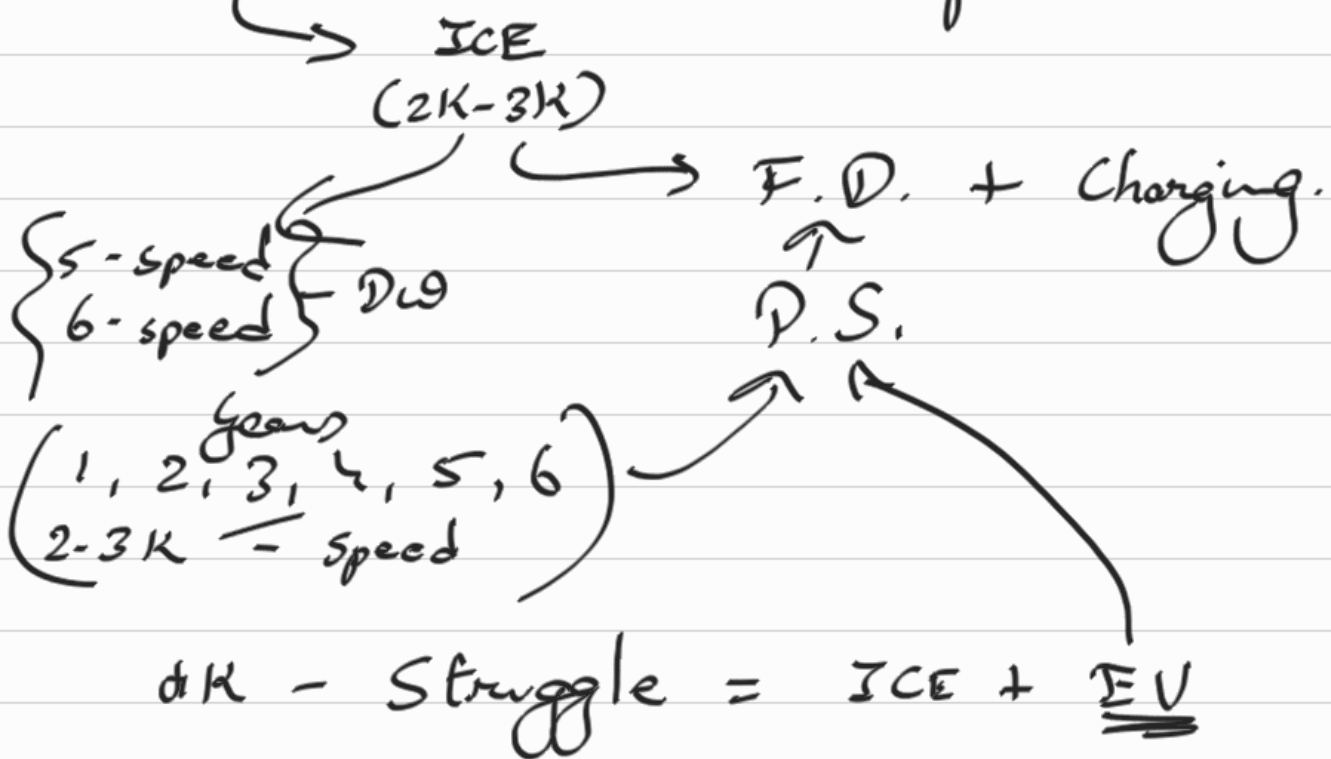
2K-3K - Petrol Max. Thermal Eff.



3. Congestion $\rightarrow$ 40 kmph - 60 kmph

ICE Eff $\uparrow$ (2K-3K rpm) + Battery $\downarrow$

4. High Speed cruising - 90 kmph max. (MIOC) Modified Indian DC



5. Deceleration / Braking

- How much to Regen.

Power Driving - Level Regen

Idle, L1, Braking (Disc Brake)

Regen $\rightarrow$ Battery

Regen First

80% 100%

Max Battery Health.

6. Max. Acc

CE + EV - Max Acc.

7. Idling - Dead stop traffic.

SoC - EV only. (SoC is sufficient for the time / distance)

SoC (not-sufficient)
 → Turn on ICE to ICE-idling
 → Per Charging

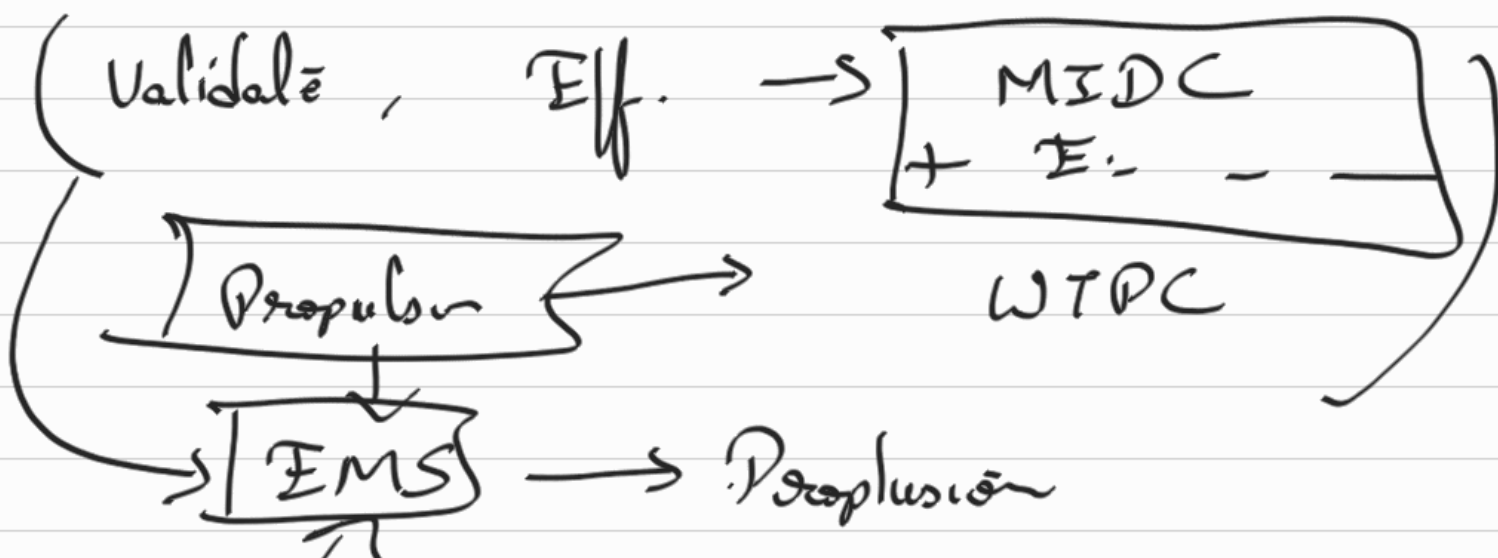
Enough for Congestion / Urban zone.

8. ICE at req. rpm → generate Elec.

Drive wheel ← e.m.

Charging ←

Series - Parallel HE



Date - Vehicle & Drive

→ Sp. grad.
 Current Acc. / De-acc.

A-class, K.W - 1800 kgs.
 ICE, Battery, MA.

